

## PROFESSIONAL SUMMARY

**Systems-oriented Engineering Leader & Venture Founder** with a 12+ year track record of scaling global technical organizations from 0 to 1 across Asia, Australia, and the Americas. Expert in transforming legacy technical debt into high-availability enterprise stacks and bridging the gap between **MIT-tier research and commercial viability**. A strategic architect of **socially responsible technology**, I have led the development of complex platforms for **renewable energy, carbon tracking, and urban mobility**. Proven success in securing **\$250k+ in non-dilutive innovation funding** and building high-retention, cross-functional cultures that align engineering excellence with meaningful environmental and social impact.

## PROFESSIONAL EXPERIENCE

### Pulse Clinic

Singapore & Bangkok, Thailand

Chief Technology Officer (CTO)

2025 – Now

\$ {C#, TypeScript, Docker, Agile, CD/CI, Terraform, GitHub Actions, Jira}

- **Engineering Transformation:** Re-engineered a decade of legacy technical debt by implementing CI/CD, Agile methodologies, and Clean Code principles.
- **Organizational Growth:** Built a cross-functional Agile team (UX/Product/Eng) by vetting hundreds of candidates to transform a developer-only silo.
- **System Reliability:** Improved platform uptime by over 90%, reducing critical outages from 3x weekly to <1x monthly.
- **Regional Leadership:** Strategized technology for 200+ employees serving tens of thousands of customers across 6 international markets (TH, MY, SG, PH, HK).
- **Digital Transformation:** Architected and executed migrating from a legacy in-house Clinic Management System to an enterprise solution with zero data incidents.

### Flo Energy

Singapore & Melbourne, Australia

Tech Lead -> Engineering Manager

2021 – 2025

\$ {Kotlin, TypeScript, GraphQL, Kubernetes, Docker, Agile, CD/CI, Terraform, Bitbucket, Jira, IoT}

- **Organizational Scaling:** Built 4 global engineering teams from zero, scaling from 4 to 20+ members in 1.5 years with established onboarding and innovation culture.
- **Product Expansion:** Oversaw 0-to-1 product development and expansion into international markets, including Renewable Energy Certificate implementations.
- **Strategic Integration:** Led high-stakes technical integrations with Singapore's Energy Market Company and drove evidence-based corporate decision-making.
- **Operational Excellence:** Hardened infrastructure by modernizing CI/CD pipelines, automated testing, and global privacy/security policies.
- **Technical Strategy:** Directed quantitative research and corporate deliberation to steer high-level technology improvements and IoT architecture.

### Singapore - MIT Alliance for Research and Technology (SMART)

Singapore  
2020 – 2021

Tech Lead Software Engineer

\$ {Node.js, Angular, Express, TypeScript, WSL, Linux, Ubuntu, ECharts, Open Layers, Python}

- **System Modernization:** Architected and engineered a system-wide UI overhaul for SimMobility's agent-based simulation platform, migrating legacy components to a high-performance Angular framework.

## Massachusetts Institute of Technology (MIT) + SMART

Boston, USA & Singapore

Project Lead Software Engineer

2014 – 2020

\$ {Android, Java, JavaScript, Mapbox API, PostgreSQL, REST, Spring, Jersey, Google Maps API, PostGIS, R, R Studio}

- **Team Leadership:** Led a team of 9 in the R&D of Flocktracker, a specialized platform for high-precision, georeferenced field surveys.
- **Full-Stack Architecture:** Architected and programmed the end-to-end cloud stack, including the RESTful back-end and the Android client-side UX.
- **Engineered Resilience:** Developed a custom asynchronous system for geospatial data collection and route tracing, optimized for unreliable offline environments.
- **Global Scale:** Directed community outreach and negotiated software deployments in 30+ locations across 4 continents.
- **Technology Commercialization:** Successfully translated research into a commercial venture by launching Fuente to take Flocktracker to market.

## ENTREPRENEURSHIP

### Fuente Tech

Founding Chief Executive Officer (CEO), Tech Lead

Singapore

2020 – 2026

\$ {Android, Java, Gradle, GitHub Actions, Docker, AWS, Mapbox, PostgreSQL, Spring, Jersey}

- **Capital Acquisition:** Raised \$200,000+ USD in non-dilutive funding from the National Science Foundation (NSF), MIT, and SMART.
- **IP Strategy:** Negotiated complex technology licensing agreements for the Flocktracker platform with MIT and the National University of Singapore (NUS).
- **System Hardening:** Architected and implemented thousands of unit and integration tests to ensure high availability for enterprise-grade deployments.
- **Cloud Infrastructure:** Orchestrated the migration of the Flocktracker stack to AWS to transition the platform from research to commercial scale.

## EDUCATION

### Massachusetts Institute of Technology (MIT)

Boston, USA

Master of City Planning

2016–2019

Urban data, mobility and sustainability

### National Autonomous University of Mexico (UNAM)

Mexico City, Mexico

Civil Engineer

2009–2012

Soil and seismic engineering

## OTHER SKILLS

### Machine Learning Algorithms:

/ AI / Data  
Science

\$ {CART, K-Means Clustering, Linear Regression, Logistic Regression, Random Forests, Text Analytics}

### Tools:

\$ {Python, R, R Studio, D3, GGPlot2}

## LANGUAGES

**Fluent** Spanish, English

**Comfortable** French

**Familiar** Russian